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Modeling Packet Loss of Low-Earth Orbit Satellite Networks

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2025 IEEE Wireless Communications and Networking Conference (WCNC) | 979-8-3503-6836-9/25/$31.00 ©2025 IEEE | DOI: 10.1109/WCNC61545.2025.10978438

***Abstract*—The growing popularity of Low-Earth Orbit (LEO) satellites, which are increasingly cheaper to manufacture and launch, has revolutionized the Internet market. Given the impact of packet loss on the quality of service (QoS) and optimization strategies of communication systems, it is of great significance to model packet loss in LEO satellite networks. The existing packet loss models considered in the literature have different assumptions and characteristics, but it remains unclear whether they are suitable for LEO satellite networks. This paper aims to evaluate several packet loss models. Through the evaluation of several metrics, the advantages and disadvantages of each model are discussed. We then introduce the Markovian Arrival Process (MAP) as a choice for packet loss modeling, and the experimental results show that the performance is better than that of existing models.**

***Index Terms*—LEO satellite networks, Markovian arrival process, Packet loss modeling, Markov chain, Generative adversarial network**

I. INTRODUCTION

With the advancement of sixth-generation (6G) mobile communication technology, the creation of a seamless, intelligent network with ubiquitous coverage has become a key challenge in the field [1]. Among the emerging solutions, Low-Earth Orbit (LEO) satellite communication systems have gained significant attention due to their unique technical advantages and promising potential for a wide range of applications [2]. The ongoing deployment of LEO satellite constellations, such as Starlink, OneWeb, and Kuiper, highlights the growing capacity of these systems to provide global broadband Internet access [3]. However, compared to traditional terrestrial cellular networks, LEO satellite communication faces challenges in terms of network reliability and quality of service (QoS). These issues stem from the dynamic nature of LEO satellites [4], including frequent satellite handovers, complex channel conditions, and high mobility.

In LEO communication systems, the limited coverage area of each satellite and its high speed result in frequent handovers as the satellite moves out of the coverage range of the ground antenna. To maintain continuous connectivity, the system periodically switches to a new satellite and reconfigures the network [5]. These frequent handovers cause dynamic changes in the network path and fluctuations in propagation delays [6], which can disrupt routing, congestion control, and loss recovery mechanisms [7], leading to significant packet loss. This packet loss is a critical factor that affects the overall end-to-end performance of LEO satellite communications, resulting in increased end-to-end communication latency and decreased throughput, and therefore impacts the user experience and overall quality of service for real-time applications [8]. Therefore, modeling packet loss behavior is essential to optimize network design and enhance the performance of LEO communication systems.

To accurately simulate LEO satellite networks, several studies have focused on developing models to describe signal transmission performance, primarily analyzing fading and interference in communication using stochastic geometic techniques [9], [10]. Other research has concentrated on the link-level performance [11]], [12], [13]. However, despite extensive research in these areas, there has been little work on modeling the end-to-end communication-level packet loss for LEO communication networks.

The goal of packet loss modeling is to develop a mathematical framework that describes the behavior of packet loss that may caused by frequent satellite handovers, fluctuations in channel conditions, and network congestion. These models may help predict packet loss, helping design schemes to improve the network stability and user experience. For example, a valid model may provide guidance for designing proper erasure correction codes for LEO networks [14].

The main contributions of this paper are as follows: we tested the capability of common models to model packet loss in LEO network scenarios. Consider the frequent random packet loss in LEO networks, we propose to model packet loss in LEO networks as a Markovian Arrival Process, and the effectiveness and advantages of Markovian Arrival Process for packet loss modeling in LEO network scenarios are analyzed through comparative experiments.

The rest of this paper is organized as follows: Section 2 reviews common packet loss models; Section 3 reviews the Markov arrival process and introduces how to use it to model the packet loss; Section 4 introduces the acquisition method of Starlink real data and gives the evaluation metrics of the model. The fitting performance of the model is evaluated and analyzed; Finally, Section 5 summarizes the paper.

II. PACKET LOSS MODELS REVIEW

Initially, Gilbert proposed a basic two-state Markov chain packet loss model (Gilbert model) [15]. Gilbert model is widely used as a loss model of loss events in networks. Although this model may be unable to accurately describe the loss behavior in real networks [16], [17], it provides foundation for the innovation and development of Markov chain based models. The Gilbert-Elliott model and 4-State Markov model [18], commonly used in network simulators such as netem [19], have not been studied whether they are applicable to LEO networks. The Loss Run-Length model [20] extends the original packet loss state of Gilbert model to several possible packet loss states, so as to model the burst packet loss that may occur in the actual network. The Receive-Loss Run-Length model [20] extends the receiving state based on the Loss Run-Length model model. Generative adversarial network [21] is a deep learning model that co-optimizes the generator and discriminator through adversarial training, ultimately enabling the generator to generate fake data similar to the real data distribution.

In the following, we denote the random process of packet loss of length S as a sequence $B = [b(1), \ldots, b(S)]$, where $b(i) = 1$ indicates packet loss and $b(i) = 0$ otherwise. We can use the inteval between the ith and $(i+1)$th loss as t_i, and hence the packet loss interval series $q = [t(1), \ldots, t(n_{loss})]$.

A. Gilbert-Elliott Model

Gilbert-Elliott model (GE) is a second-order hidden Markov chain model [15]. It assumes that packet loss occurs in both states of the model (Fig. 1). The GE model assigns a packet loss probability P_r to the receiving state R, and assigns a packet loss probability P_l to the packet loss state L.

Define the coefficient $a_{ij}, i, j \in \{0, 1\}$, the calculation formulas are as follows:

$$a_{i0} = \frac{P(ii)^2 - P(i)P(iii)}{P(11) - P(1)^2}, a_{i1} = \frac{P(iii) - P(i)P(ii)}{P(11) - P(1)^2} \quad (1)$$

where $i \in \{0, 1\}$, if $i = 1$, $P(i) = P(1)$ is the probability that 1 occurs in data B, and so on for $P(ii)$ and $P(iii)$. Based on a_{ij}, we can get the following calculation formulas:

$$P_L = \frac{(a_{11} + a_{01}) P_r - a_{11}}{(P_r - P_l)} \quad (2)$$

$$P_R = 1 - \frac{a_{11} - (a_{11} + a_{01}) P_l}{(P_r - P_l)} \quad (3)$$

$$P_l = \frac{1}{2}\left(\alpha + \sqrt{\alpha^2 - 4\beta}\right), P_r = \frac{1}{2}\left(\alpha - \sqrt{\alpha^2 - 4\beta}\right) \quad (4)$$

where

$$\alpha = \frac{a_{11} + a_{01} + a_{00} - a_{10} - 1}{a_{11} + a_{01} - 1}, \beta = -\frac{a_{10}}{a_{11} + a_{01} - 1} \quad (5)$$

B. 4-State Markov Model

Salsano et al. proposed a 4-State Markov model (4SM) with two good and bad states [18] as shown in Fig. 2. This model emulates both burst and gap periods to better describe how real network loss events happen. State S_1 represents packet received successfully, state S_2 represents packet received with a burst, state S_3 represents packet lost within a burst, state S_4 represents isolated packet lost within a gap.

The 4SM model has been implemented in Linux's netem emulator, and has been widely used to simulate packet losses for computer networks. When in use [18], it is often assumed $p_{41} = 1$. Solving the system, we get the states probabilities:

$$\begin{cases} \pi_1 (p_{13} + p_{14}) = \pi_3 p_{31} + \pi_4 \\ \pi_3 p_{32} = \pi_2 p_{23} \\ \pi_1 p_{14} = \pi_4 \\ \pi_1 + \pi_2 + \pi_3 + \pi_4 = 1 \end{cases} \quad (6)$$

C. Loss Run-Length Model

In [20], Sanneck and Carle proposed an extension of Gilbert model, called Loss Run-Length model (LRL), as shown in Fig. 3. The LRL model extends the original loss state of Gilbert model to N possible loss states. P_R is the probability of moving from the receiving state (R) to the first packet loss

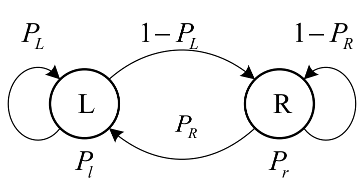

Fig. 1. Gilbert-Elliott Model

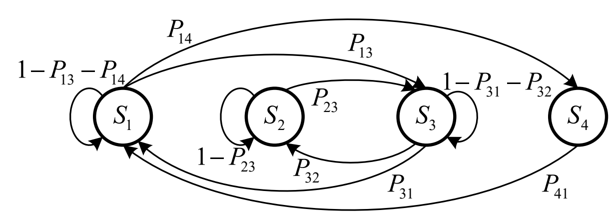

Fig. 2. 4-state Markov Model

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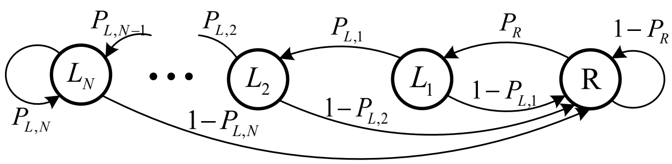

Fig. 3. Loss Run-Length Model

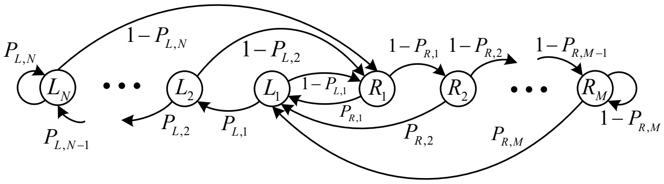

Fig. 4. Receive-Loss Run-Length Model

state (L_1), where $P_{L,k}, k = 1, \ldots, N-1$ is the transition probability from state L_k to state L_{k+1}.

Let $V_0 = [v_0(1), \ldots, v_0(m)]$ and $V_1 = [v_1(1), \ldots, v_1(n)]$ be the two vectors of successive lengths of 0s and 1s in B, m indicates that there are m consecutive 0's in B, and n indicates that there are n consecutive 1's in B, where $v_j(i) = k, j = 0, 1$ represents the i th consecutive 0s (1s) of length k in B. Let $C_j = [c_j(1), \ldots, c_j(S_j)]$ be the sorted array of V_j occurrences in the record, where $S_j = \max(V_j), j = 0, 1$. Consider the inverse cumulation and vector, expressed as $G_j = [g_j(1), \ldots, g_j(S_j)]$, $g_j(k)$ is calculated as:

$$g_j(k) = \sum_{i=k}^{S_j} c_j(i), j = 0, 1 \tag{7}$$

The transition probability formula of the LRL model can be determined from data B as:

$$P_R = f_B, \tag{8}$$

$$P_{L,k} = \frac{g_1(k+1)}{g_1(k)} \tag{9}$$

where f_B is the ratio of the length of V_0 to S.

D. Receive-Loss Run-Length Model

Sanneck and Carle also proposed the Receive-Loss Run-Length model (RLRL) based on LRL model [20], and the description of data burst reception is added. As shown in Fig. 4, the loss state of the RLRL model is consistent with that of the LRL model, and the receiving state expands to M possible receiving states. $P_{R,1}$ is the probability of moving from the first received state (R) to the first lost packet state (L), where $P_{R,k}, k = 1, \ldots, M$ represents the probability of receiving $k+1$ packets in a row. $P_{L,k}$ is defined in (9), $P_{R,k}$ is defined as:

$$P_{R,k} = \frac{g_0(k+1)}{g_0(k)} \tag{10}$$

E. Generative Adversarial Network Model

Generative adversarial network (GAN) [21] is a powerful generative model whose core ability lies in its ability to capture and learn the potential distribution in a data set. The basic framework of a GAN consists of two main components: Generator (G) and Discriminator (D).

The training process of GAN involves the adversarial training of G and D. First, the weights of both G and D are initialized to random values [22]. In each training epoch, we first randomly extract a batch of real data x from the real data distribution q, and randomly extract a batch of noise vector z from the noise distribution $p_z(z)$, and generate false data $G_z(z)$ through G. Then, calculate the loss function of D

$$\begin{aligned} L_D = & -E_{x \sim q_{\text{data}}}(x)[\log D(x)] \\ & -E_{z \sim p_z}(z)[\log(1 - D(G(z)))] \end{aligned} \tag{11}$$

The weights of D are updated by back propagation. A number of noise vectors z are extracted from the noise distribution $p_z(z)$ again to generate false data $G_z(z)$ and calculate the loss function of G

$$L_G = -E_{z \sim p_z}(z)[\log(D(G(z)))] \tag{12}$$

The weights of G are updated by back propagation. After training, G is able to generate high-quality fake data that is statistically very close to the real data distribution q.

III. MARKOVIAN ARRIVAL PROCESS MODEL

Markovian Arrival Process (MAP) was proposed in [23] and is a generalization of Continuous Time Markov Chains (CTMC) [24]. MAP is widely used in stochastic modeling for Internet traffic [25]. It is a combination of Markov chains and point processes on countable state spaces, which can capture complex dependencies of arrival events.

A. Model Introduction

A MAP is usually defined by two matrices, $\boldsymbol{D_0}$ and $\boldsymbol{D_1}$. Both $\boldsymbol{D_0}$ and $\boldsymbol{D_1}$ are matrices of $m \times m$, where m is the number of states of the Markov chain. $\boldsymbol{D_0}$ reflecting phase transitions without packet arrivals, $\boldsymbol{D_1}$ corresponding to arrival rates in the m states. $\boldsymbol{D_0}$ contains negative diagonal elements and nonnegative off-diagonal elements, while all the elements of the $\boldsymbol{D_1}$ are non-negative. The $\boldsymbol{D_0}$ and $\boldsymbol{D_1}$ matrix of an M-order MAP can be expressed as:

$$\boldsymbol{D}_0 = \begin{pmatrix} -\mu_{1,1} & \mu_{1,2} & \cdots & \mu_{1,m} \\ \mu_{2,1} & -\mu_{2,2} & \cdots & \mu_{2,m} \\ \vdots & \vdots & \ddots & \vdots \\ \mu_{m,1} & \mu_{m,2} & \cdots & -\mu_{m,m} \end{pmatrix} \tag{13}$$

$$\boldsymbol{D}_1 = \begin{pmatrix} \lambda_{1,1} & \lambda_{1,2} & \cdots & \lambda_{1,m} \\ \lambda_{2,1} & \lambda_{2,2} & \cdots & \lambda_{2,m} \\ \vdots & \vdots & \ddots & \vdots \\ \lambda_{m,1} & \lambda_{m,2} & \cdots & \lambda_{m,m} \end{pmatrix} \tag{14}$$

where $\mu_{i,i} = \sum_{j=1, j \neq i}^{m} \mu_{i,j} + \sum_{j=1}^{m} \lambda_{i,j}$. Thus, $\boldsymbol{D} = \boldsymbol{D_0} + \boldsymbol{D_1}$ is an infinitesimal generator for the underlying Markov chain. The fitting MAP model based on arrival time interval data can be achieved via moment matching (MM) or expectation maximization (EM).

Algorithm 1: MAP Fitting Using the EM Algorithm

```
Data: data q, the order of the generated matrix N
Result: {D_0, D_1}
Initialize max iterations, stop condition;
if orders is a single integer then
    Test possible branch number-order combinations;
end
Calculate and initialize the EM algorithm initial
  parameters and generator matrix;
while stopCond is not met and steps ≤ maxIter do
    E-step:;
        Calculate density for each branch;
        Update forward and backward variables;
    M-step:;
        Update iteration parameters and transition
          probability matrix P;
end
return {D_0, D_1}.
```

B. Packet Loss Model Construction

Using MAP to model packet loss means that we regard packet loss as a arrival process. To construct $\{\boldsymbol{D_0}, \boldsymbol{D_1}\}$, we record the interval between packet loss, which is the packet loss interval series q described earlier. Using the MM or EM fitting algorithm, we can get the $\{\boldsymbol{D_0}, \boldsymbol{D_1}\}$ matrix from the data q. Algorithm 1 outlines how to obtain the $\{\boldsymbol{D_0}, \boldsymbol{D_1}\}$ matrix describing the packet loss arrival process from using data q and a fast EM algorithm [26], [27].

IV. Model Evaluation and Analysis

A. Experimental Data Acquisition

At present, Starlink is one of the main source from which we can obtain the real-life end-to-end data over LEO satellite network. In this paper, a measured system as shown in Fig. 5, where the Starlink user terminal is set up in Victoria, British Columbia, Canada, and the peer server is set up in data centers near the associated home Point-of-Presence (PoP) of the user terminal in Seattle, USA. This setup enables the measured data to avoid interference by other ground segment services to the greatest extent. By running IRTT [28] on the local host and the remote server, packets are continuously sent between the server and the local host within 12 hours. The packet send interval is 10 ms, and we chose downlink data for the experiment, because downlink packet loss is much more than uplink packet loss, which is more suitable for the modeling of packet loss characteristics.

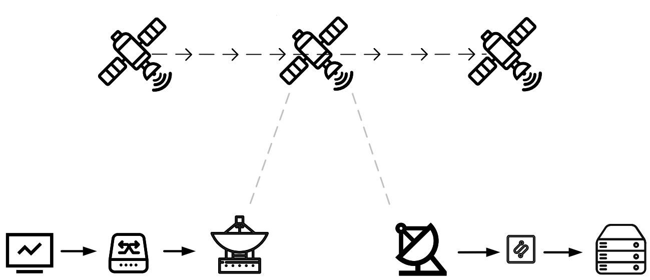

Fig. 5. Measurement system of Starlink network

After processing the measured data, two sets of correlation sequences were obtained: packet loss sequence B and packet loss interval series q. Data B is mainly used for modeling and generating Markov chain model parameters. Data q is mainly used for GAN model training and MAP matrix $\{\boldsymbol{D_0}, \boldsymbol{D_1}\}$ generation. With an iRTT transmission interval of 10 ms, data B with a length of about 4.5 million and data q with a length of about 60,000 can be obtained after each 12-hour measurement. We continuously recorded the packets data from September 21 to September 25, 2024.

B. Evaluation Metrics

To measure the accuracy of the modeling, we consider the following performance indicators:

- Gap distribution: Gap (G) is defined as a continuous 0's between two 1's in B. The gap distribution $G(m_g)$ is defined as the probability distribution function for gaps of length m_g.
- Error cluster distribution: Error cluster (EC) is defined as consecutive 1's between two 0s in B [29]. The error cluster distribution $P(m_c)$ defines the probability distribution function for the occurrence of error clusters of length m_c or more.
- Error burst distribution: Error burst (EB) is defined as a sequence of error clusters with gaps greater than or equal to η length on both sides, or with gaps smaller than η between them. Where η is a predefined positive integer [30], [31]. The error burst distribution $P_{EB}(m_e)$ is the probability distribution function of the error burst of length m_e.
- Error-free burst distribution: Error-free burst (EFB) is defined as a gap of length greater than or equal to η. The error-free burst distribution $P_{EFB}(m_e)$ is defined as the probability distribution function of an error-free burst of length m_e.

The above items can be illustrated by referring to an example sequence given in Fig. 6, where η is 3.

$$\cdots\ 1\ \overbrace{\underbrace{0\ 0\ 0\ 0}_{\text{G}}}^{\text{EFB}}\ \overbrace{\underbrace{1\ 1\ 1}_{\text{EC}}\ \underbrace{0}_{\text{G}}\ \underbrace{1\ 1\ 1\ 1}_{\text{EC}}}^{\text{EB}}\ \overbrace{\underbrace{0\ 0\ 0}_{\text{G}}}^{\text{EFB}}\ 1\ \cdots$$

Fig. 6. An example of packet loss sequence

C. Model Comparison

In the recorded downlink data, we selected two days of frequent packet loss period data (18-22h) for the experiment, and the data packet loss characteristics were similar. Specifically, the 18-22h data of September 21 (22) is used as the training

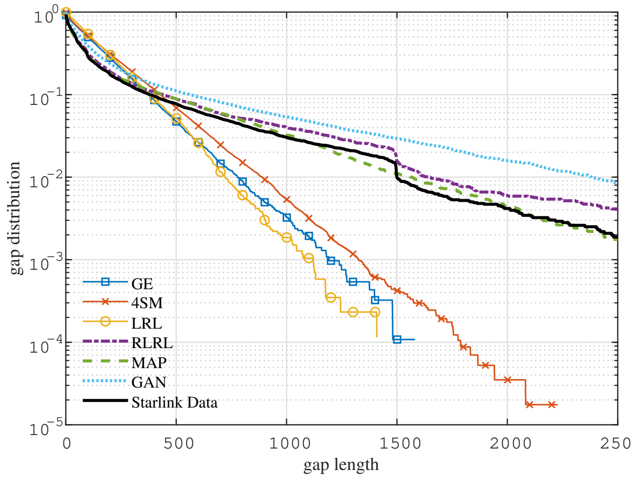

Fig. 7. Gap distribution

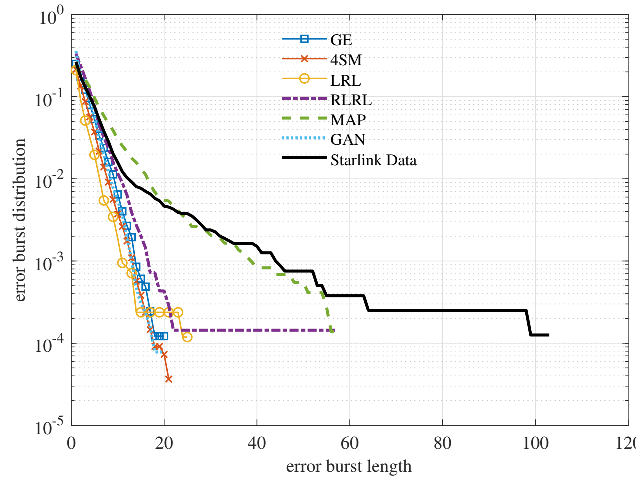

Fig. 9. Error burst distribution

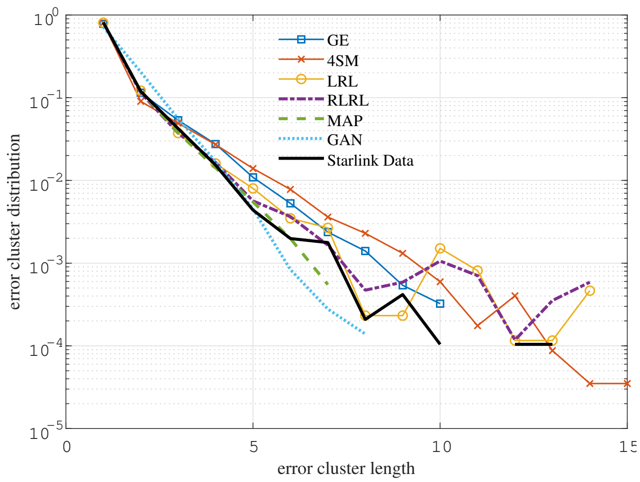

Fig. 8. Error cluster distribution

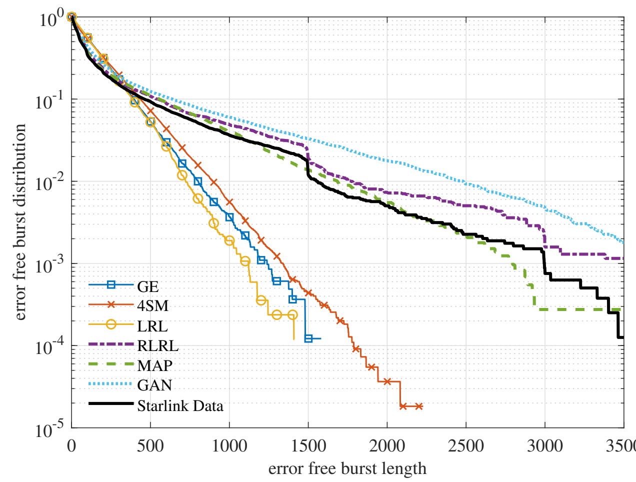

Fig. 10. Error-free burst distribution

(test) set. We use BuTools [32] to generate the $\{\boldsymbol{D_0}, \boldsymbol{D_1}\}$ matrix of the MAP. Under the above evaluation metrics, the models are evaluated by comparing the generated data from the fitted models with the test data. The measurement data of Sept. 21 and 22, and the model fitting script are available at https://github.com/yeliqseu/starlink-loss-modeling.

The GE and 4SM models only perform well in Fig. 8. In the other three metrics, their performance is far from the test data. This means that the packet loss data generated by them cannot show the packet loss characteristics of LEO networks.

As can be seen from Fig. 7 and 10, the LRL model fails to model the feature distribution of received events in the network, while RLRL model and GAN model can approximate the real data. According to Fig. 8 and 9, LRL, RLRL and GAN models can generate continuous packet loss close to the real data, but the packet loss burst distribution of these modelss is much different from that of LEO network.

TABLE I
NUMBER OF MODEL STATES

Model	LRL	RLRL	MAP
State	59	4842	8

In each of the four assessment indicators, MAP achieves a good performance, and only the curve of the MAP model is close to the real data in Fig. 9. This shows that MAP can better model the packet loss characteristics of LEO networks, and also proves that modeling packet loss as arrival process is a feasible modeling scheme. As shown in Table 1, MAP has a more concise mathematical structure. The number of states of the MAP model can be self-specified before the experiment, while the number of states of the LRL and RLRL models is determined by the data B. This facilitates subsequent delay performance analysis and forward error correction design.

V. CONCLUSIONS AND FUTURE WORK

In this paper, a variety of packet loss models are evaluated for LEO satellite network packet loss modeling. Through experiments, we find that the GE model and 4SM model that are commonly used in network simulators such as netem [19] cannot accurately model packet loss in LEO networks. LRL, RLRL and GAN models can capture part of the packet loss characteristics, but they are still insufficient to represent the packet loss characteristics of LEO networks. In the complex packet loss scenario of LEO networks, MAP model performs better than most common packet loss models.

At present, the modeling work of LEO satellite packet loss still needs a lot of experiments and more model exploration attempts. In the future work, we will further consider the relationship between satellite switching period, packet loss delay information and packet loss in LEO networks, and try to mine a model that is more close to the characteristics of packet loss in LEO network. Around the idea of modeling packet loss as arrival process in LEO network, the batch Markovian arrival process is also a modeling scheme worth exploring.

ACKNOWLEDGMENT

This work was supported by the National Natural Science Foundation of China (62341104) and the Postgraduate Research & Practice Innovation Program of School of Information Science and Technology, Nantong University (NTU-SISTPR24 09).

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